ReproChem · Workflow or agent? Five real requests

Agentic AI · PGE 5 · aivancity · Session 1 lab, agent sheet · 4 November 2026. ReproChem extracts the 7 fields of a computational-chemistry protocol (functional, basis set, dispersion, solvation, charge, multiplicity, software) and compiles them into runnable calculations.

#	Request	Decision	Justification (one sentence)	Safeguards
1	Read the Computational Details of this PDF and give me the 7 fields (functional, basis set, dispersion, solvation, charge, multiplicity, software) as JSON.	**Workflow** prompt chaining	The path never changes (find the methods section, extract into a strict schema, flag empty fields), so code fixes the steps and the gold-annotated papers check the output.	none
2	Turn this protocol (wB97X-D/def2-TZVP, SMD water, charge 0, singlet, Gaussian 16) into a ready-to-run input file for my geometry.	**Workflow** single compile step + validator	Compilation is a deterministic template plus a validator (for example, no extra D3 keyword, since wB97X-D already contains its dispersion term), so a loop would only add cost.	none
3	For these 20 papers on conformer energies, tell me which ones report all 7 fields and which ones omit charge, multiplicity or solvation.	**Workflow** parallelisation	It is the same extraction mapped over 20 independent papers and then aggregated, which is the parallelisation pattern with a known number of calls.	none
4	The paper generated conformers with CREST (GFN2-xTB) and re-optimised the lowest ones at B3LYP-D3(BJ)/def2-SVP; reproduce the ensemble for my molecule and tell me which conformer is the global minimum at their level.	**Agent** agent loop with approval	Key details such as the energy window and how many conformers were re-optimised are often missing, so the agent must search the SI, decide, run, check convergence and imaginary frequencies and retry, with my approval before any cluster job is submitted.	human approval before write or costly actions
5	My relative conformer energies differ from Table 2 of the paper by up to 2 kcal/mol; find out why.	**Agent** read-only diagnosis loop	The next check depends on the last result (D3 vs D3(BJ), solvation model, geometry level, electronic vs Gibbs energies, temperature), and since it only reads files a wrong step is cheap.	none

Checklist from the S1 slide "When not to build an agent"

#	Are the steps unknown until you see the task?	Does it need fresh data or actions in other systems?	Is a wrong step cheap and easy to undo?	Can you afford 10x the cost and latency of one call?	Can you check success automatically?	Checklist says
1	no	yes	yes	yes	yes	workflow (agrees)
2	no	yes	yes	yes	yes	workflow (agrees)
3	no	yes	yes	yes	yes	workflow (agrees)
4	yes	yes	no	yes	yes	agent (agrees)
5	yes	yes	yes	yes	yes	agent (agrees)

Rule applied: fixed steps lead to a workflow; steps unknown in advance and an affordable budget lead to an agent loop; a costly or irreversible step adds human approval. Generated with workflow_or_agent.py from reprochem_requests.py.